Arij Asad

Curriculum Vitae

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Research Profile

Final-year PhD researcher in applied mathematics at the University of Bristol, studying buoyancy-driven mixing and plume-generated stratification in confined environments, with applications to nuclear-waste storage. Methods include Planar Laser-Induced Fluorescence (PLIF), mathematical modelling, similarity analysis and finite-element simulation.

Research interests: buoyant plumes, mixing and stratification; experimental fluid dynamics; molecular processes in fluids; and computational fluid dynamics and finite-element methods.

Education

University of Bristol November 2022 – present

Bristol, United Kingdom

PhD in Mathematics

Faculty of Science and Engineering; School of Electrical, Electronic and Mechanical Engineering

- Supervisors: Dr Andrew Lawrie and Dr Stuart Thomson.
- Thesis: *Mixing behaviour of plumes.*

University of Cambridge October 2019 – June 2022

Cambridge, United Kingdom

BA (Cantab), Mathematics

- MA (Cantab), conferred April 2026.

Research and Professional Experience

MSc Research Project Support: Pengzhu Shi January 2025 – September 2026

University of Bristol

Provided day-to-day research guidance on *Interaction of twin buoyant plumes in a confined environment.*

Widening Participation Tutor July 2023 – present

Faculty of Science and Engineering, University of Bristol

She Talks Science: STEP Summer School July 2023 – August 2025

Murray Edwards College, University of Cambridge

- Director, October 2024 – August 2025: created teaching materials and timetables; made admissions decisions; delivered classes and supervisions; and ran evening activities.
- Deputy Director, July 2023 – August 2024: contributed to teaching materials; delivered classes and supervisions; and ran evening activities.

Cambridge Mathematics Open Intern July 2022 – August 2022

Department of Applied Mathematics and Theoretical Physics, University of Cambridge

- Project: *The motion of anisotropic carbon nanoparticles in shear flow: a hydrodynamic perspective.*
- Supervisor: Dr Catherine Kamal.

Research Outputs

Manuscripts in preparation

- **Arij Asad**, Andrew G. W. Lawrie and Stuart J. Thomson, *On the mixing behaviour of plumes in a confined environment.*
- **Arij Asad**, Andrew G. W. Lawrie and Stuart J. Thomson, *Stratification development in a laminar crossflow.*
- **Arij Asad**, Andrew G. W. Lawrie and Stuart J. Thomson, *Stratification development in a confined circular domain with and without laminar crossflow.*

Research software

- PLIF: MATLAB tools for processing Planar Laser-Induced Fluorescence images, extracting density fields and analysing plume-driven stratification experiments.

Selected Presentations and Posters

Osborne Reynolds Day 2026
On the mixing behaviour of plumes in a confined environment

British Applied Mathematics Colloquium (BAMC), University of East Anglia, Norwich 2026
On the mixing behaviour of plumes in a confined environment (poster)

NWS RSO Annual Conference, Manchester 2026
How does hydrogen build up in a closed vault environment? (poster)

2nd European Fluid Dynamics Conference (EFDC2), Dublin 2025
Mixing behaviour of plumes

NWS RSO Annual Conference, Bristol 2025
Hydrogen ventilation in a GDF (poster)

NWS RSO Annual Conference, Sheffield 2024
Mixing behaviour of laminar plumes with and without a ventilating crossflow

NWS RSO Advanced Manufacturing Webinar (online) October 2023
Ventilation of hydrogen in UILW vaults (recording)

NWS RSO Annual Conference, Leeds 2023
Hydrogen ventilation in a geological disposal facility

Cambridge Mathematics Placements (CMP) Presentation Day 2022
The motion of anisotropic carbon nanoparticles in shear flow: a hydrodynamic perspective

Examining and Admissions

Marking Supervisor 2026
OCR (Oxford Cambridge and RSA)

- STEP Mathematics 2.
- Led a team of markers; supported consistent application of the mark scheme; and marked and moderated scripts.

STEP Marker 2024 – present
OCR (Oxford Cambridge and RSA)

- STEP Mathematics 2 and STEP Mathematics 3.
- Applied mark schemes and provided feedback on their implementation.

Mathematics Undergraduate Admissions Interviewer 2024
Homerton College, University of Cambridge
Developed interview questions; interviewed and assessed applicants.

Assessment Associate 2024
Pearson VUE
Reviewed TMUA and ESAT assessment content for scope, difficulty and clarity, and provided feedback.

Marker 2024 – 2025
University of Bristol

- Applied Analysis B and Probability and Statistics, 2025.
- Applied Linear Algebra Coursework, Engineering Mathematics and Introduction to Pure Mathematics, 2024.

Vetter 2019
United Kingdom Mathematics Trust
Assessed draft UK Mathematical Olympiad for Girls papers for content, scope, difficulty and clarity, and provided feedback.

Marker October 2019
United Kingdom Mathematics Trust
UK Mathematical Olympiad for Girls.

Teaching Experience

Supervisor Michaelmas 2024 – present

University of Cambridge, across ten colleges

Part II Fluid Dynamics II.

Lead Demonstrator September 2024 – March 2025

University of Bristol

Wrote and delivered weekly tutorials.

- Probability and Statistics, January 2025 – March 2025.
- Introduction to Pure Mathematics, September 2024 – December 2024.

Teaching Assistant January 2024 – January 2026

University of Bristol

- Engineering Mathematics, September 2024 – January 2026.
- Modelling in Applied Mathematics, September 2024 – December 2024.
- Applied Linear Algebra, March 2024 – April 2024.
- Continuum Mechanics, January 2024 – May 2024.

Private Mathematics Tutor October 2019 – present

Primarily online; occasional in-person teaching

- A-level Mathematics, Further Mathematics, Physics and Computer Science, October 2019 – present.
- STEP and TMUA preparation, October 2020 – present.
- MAT preparation, 2020 – 2025.

Outreach and Service

Specialist Mathematics Tutor July 2026

Dipont Education Summer Camp, Wuxi

Outreach Speaker July 2024 – July 2026

University of Bristol

Delivered research talks for widening-participation programmes. Insight into Bristol Summer School, July 2024, July 2025 and July 2026; Next Step Bristol, November 2025; Virtual Summer School, May 2025 and May 2026; and Sutton Trust Summer School, July 2024.

Summer School Teaching 2019 – 2022

University of Cambridge

Delivered classes and supervisions and ran evening activities for the Murray Edwards STEP Summer School, 2019 – 2022; the Sutton Trust Summer School, 2021 – 2022; and Intro to Maths at Uni, July 2021.

Speaker 2019

Murray Edwards College Maths Circle, Cambridge

Sphere packing.

Panellist

- Blue Education Oxbridge Interview Seminar, October 2025.
- Mathematics Open Day, University of Cambridge, 2022.
- Sutton Trust Experience Cambridge Day, 2020 – 2022.

Technical Skills and Languages

- Programming and technical computing: MATLAB, Python and C; scientific typesetting in LaTeX.
- Experimental methods: PLIF image acquisition and MATLAB-based image processing.
- Numerical methods: finite-element modelling in FEniCS, including incompressible-flow and advection–diffusion simulations.
- High-performance computing: running and managing parallel simulations on the University of Bristol's BlueCrystal and BluePebble systems.
- Urdu: reading and conversational proficiency.